

Krishna Gupta

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PROFESSIONAL SUMMARY

Innovative AI/ML Engineer and B.Tech in Data Science student (Class of 2027) with dual industry internship experience architecting Large Language Model (LLM) post-training workflows, autonomous multi-agent systems (LangGraph), and high-throughput data pipelines. Proven expertise across the entire MLflow lifecycle (tracing, evaluation, prompt templates, AI gateways, agent servers, hyperparameter tuning), computer vision (OpenCV/OCR), and full-stack cloud microservices with 99.9% uptime. Certified AWS Developer with deep technical proficiency in PyTorch, Scikit-learn, FastAPI, and SQL database optimization, delivering production-grade AI solutions with measurable business impact.

EDUCATION

Oriental Institute of Science and Technology

Bachelor of Technology (B.Tech) in Data Science (Class of 2027)

Bhopal, Madhya Pradesh, India

Aug 2023 – May 2027

TECHNICAL SKILLS

Generative AI & Agentic Systems: Large Language Models (LLMs), Vision-Language Models (VLMs), LangGraph, LangChain, Autonomous Multi-Agents, Retrieval-Augmented Generation (Hybrid RAG), Prompt Engineering, Safety Guardrails, Model Alignment (RLHF/DPO), Vector Databases (FAISS, pgvector), Hugging Face Transformers

MLOps & MLflow Ecosystem: MLflow (Tracing, Model Evaluation, Prompt Templates, AI Gateways, Agent Server, Scikit-Learn MLOps, PyTorch MLOps, Hyperparameter Tuning), Experiment Tracking, Model Registry, CI/CD Pipelines, Docker Containers, AWS Cloud Infrastructure

Machine Learning & Deep Learning: Supervised & Unsupervised Learning, Classification, Regression, Time-Series Forecasting (LSTM), Quantitative Financial Modeling, Computer Vision (OpenCV, EasyOCR), Deep Neural Networks (CNNs), PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy

Backend, APIs & Databases: Python (Advanced), SQL (PostgreSQL, MySQL), FastAPI, RESTful APIs, Microservices, Node.js, React.js, ETL/ELT Pipelines, Dimensional Modeling (Star/Snowflake Schemas), Git, Linux Server Administration, Agile/SDLC

WORK EXPERIENCE

Ethara.AI

LLM & AI Post Training Intern (Paid Internship)

Remote

Feb 2026 – May 2026

- **Benchmarked** and evaluated 50,000+ Large Language Model outputs within LLMops workflows, boosting model alignment, precision, and safety guardrails by 15% across diverse prompt distributions.
- **Automated** Python data preprocessing, feature extraction, and validation pipelines, saving 20+ hours per week in manual quality assurance and dataset screening tasks.
- **Formulated** structured technical capability matrices and evaluation rubrics to ensure 100% compliance with enterprise data security and privacy standards.
- **Collaborated** with AI research teams to analyze edge cases, debug reward-signal discrepancies, and streamline dataset curation for downstream reinforcement learning environments.

Kanchan Pvt Ltd – Web Development Wing (Sapphire)

Full Stack Development Intern (Paid Internship)

Remote

Oct 2025 – Feb 2026

- **Architected** and deployed full-stack web infrastructure for review analytics platform “RevU Social” (revu.social) using React.js and Node.js REST APIs, driving a 20% user increase.
- **Optimized** PostgreSQL and MySQL relational database schemas and indexing strategies, reducing query latency and accelerating transaction speeds by 25%.
- **Implemented** secure JWT authentication protocols and deployed Docker container microservices on Linux cloud servers, maintaining 99.9% uptime and zero security incidents.

KEY TECHNICAL PROJECTS

KALKI 1.5 – Autonomous Multi-Agent OS & MLOps Platform

2024 – Present

Tech: Python, FastAPI, LangGraph, LangChain, LLMs, VLMs, MLflow Tracing/Evaluation, Hybrid RAG, FAISS

Live App: kalki.hg497kg.workers.dev | GitHub: [kgupta171025.github.io/KALKI-1.5](https://github.com/kgupta171025/KALKI-1.5)

- **Engineered** an autonomous multi-agent operating system integrating LLMs and Vision-Language Models (VLMs) via LangGraph and LangChain to automate complex multi-step reasoning workflows.
- **Constructed** a low-latency Hybrid RAG pipeline integrating dense vector embeddings and FAISS, cutting semantic search query retrieval latency by 40%.
- **Integrated** MLflow tracing, prompt evaluation templates, and automated guardrail logging to monitor agent communication and ensure execution safety.

Score Vision - AI (FOG Tech) – Computer Vision & Document OCR Extraction Platform

2025 – Present

Tech: Python, OpenCV, EasyOCR, Sentence-Transformers, FAISS, FastAPI, React, Docker

Live App: scorevision-ai.hg497kg.workers.dev | GitHub: [kgupta171025.github.io/FOG_Technologies](https://github.com/kgupta171025.github.io/FOG_Technologies)

- **Developed** an intelligent OCR and computer vision pipeline achieving 92% text extraction accuracy and sub-200ms semantic matching on complex imagery.
- **Integrated** Sentence-Transformers and FAISS vector databases to construct a semantic matching and ETL transformation service deployed with FastAPI.

Razorpay Recovery – Smart Payment Recovery & FinTech Transaction Pipeline

2025

Tech: Python, SQL, PostgreSQL, REST APIs, Webhooks, Data Pipelines, Transaction Analytics

Live App: razor-recovery.hg497kg.workers.dev | GitHub: [kgupta171025.github.io/Razor-Recovery](https://github.com/kgupta171025.github.io/Razor-Recovery)

- **Constructed** an automated transaction recovery pipeline capturing failed payment webhooks in real time, successfully recovering 18% of abandoned checkouts.
- **Designed** analytical SQL schemas in PostgreSQL to aggregate transaction failure patterns, automate notification triggers, and generate recovery analytics.

SHELF-SCANNER – High-Speed Computer Vision & Document OCR Platform

2025 – Present

Tech: Python, OpenCV, EasyOCR, Sentence-Transformers, FAISS, FastAPI, Cloudflare Workers

Live App: shelf-scanner.hg497kg.workers.dev | GitHub: [kgupta171025.github.io/SHELF-SCANNER](https://github.com/kgupta171025.github.io/SHELF-SCANNER)

- **Engineered** an image-to-text OCR and semantic book-matching platform using EasyOCR and FAISS, achieving sub-200ms query retrieval latency.

Future Stock Market Prediction & Quantitative Financial Modeling

2024 – 2025

Tech: Python, Scikit-learn, PyTorch MLOps, MLflow Hyperparameter Tuning, Time-Series (LSTM)

Live App: smf-ai.hg497kg.workers.dev | GitHub: [github.com/KGupta171025](https://github.com/kgupta171025)

- **Built** a quantitative time-series forecasting model using PyTorch (LSTM) and Scikit-learn to predict financial asset price trends with 88% precision.
- **Implemented** MLflow hyperparameter tuning and experiment tracking to systematically evaluate learning rates, sequence lengths, and optimizer parameters.

RevU Social – Full-Stack Review & Analytics Platform

Oct 2025 – Feb 2026

Tech: React.js, Node.js, REST APIs, MySQL, PostgreSQL, Star Schema

revu.social

- **Designed** dimensional star-schema database models in MySQL/PostgreSQL to power real-time business intelligence and performance dashboards.

CERTIFICATIONS

- **AWS Certified Developer Associate** (Infosys, Jun 2026) **Machine Learning with Python** (IBM SkillsBuild, Jun 2026)
- **Data Science & Analytics** (HP LIFE, Jun 2026) **Neural Networks and Convolutional Neural Networks** (LinkedIn, Jun 2026)
- **Deloitte Australia Data Analytics Job Simulation** (Forage, Sep 2025) **Tata Data Visualisation** (Forage, Sep 2025)

RELEVANT COURSEWORK

Data Structures & Algorithms • Database Management Systems • Operating Systems • Computer Networks • Machine Learning • Deep Learning & NLP • Big Data Engineering • Software Engineering • System Design